

## CYBER SECURITY COURSE

# Windows Malware Behavior and Initial Access

## Professional Case Report 04

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|------------------------|-------------------------------------|
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| <b>Report date</b>     | 24. septembar 2026                  |
| <b>Case ID</b>         | CSC-2026-04                         |
| <b>Discipline</b>      | Malware analysis                    |
| <b>Overall risk</b>    | Critical                            |
| <b>Evidence status</b> | Confirmed by saved Redline evidence |

### Purpose

Explain malware behavior, identify command-and-control indicators and reconstruct the likely initial infection path.

### Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

## Executive Summary

The investigation moved from static artifacts to malicious services, active network connections, browser history and process-memory strings. The combined evidence describes a complete attack chain from likely email delivery to execution, persistence, command-and-control and credential-oriented post-exploitation.

## Case Profile

|             |                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Scenario    | The investigation moved from static artifacts to malicious services, active network connections, browser history and process-memory strings. |
| Environment | Windows Redline collection with services, event logs, ports, browser artifacts and process strings.                                          |
| Tools       | Mandiant Redline, Windows Event ID 7045, Process and port analysis, Browser history, Memory strings                                          |

## Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Reviewed service installation events and matched them to current Windows services.
2. Inspected established connections and remote ownership information.
3. Reviewed browser URL and download history for the incident date.
4. Searched process strings for email and archive evidence.

## Technical Findings

| ID  | Finding                              | Risk     | Evidence and analysis                                                                                            |
|-----|--------------------------------------|----------|------------------------------------------------------------------------------------------------------------------|
| F01 | Malicious service installation       | Critical | Service creation evidence indicated persistence with SYSTEM-level execution.                                     |
| F02 | Port 4444 activity                   | High     | The non-standard listener is commonly associated with remote shell and Meterpreter-style payloads.               |
| F03 | Email-delivered archive              | Critical | Browser and memory artifacts supported a phishing-style initial access path involving a downloaded archive.      |
| F04 | Credential and remote access tooling | Critical | Earlier artifacts included Mimikatz, PsExec and FreeRDP capabilities consistent with post-exploitation activity. |

## Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

Event Log filter: Event ID 7045

Ports filter: State equals ESTABLISHED

Process Strings search using email page title

## Risk Assessment

| Risk                     | Level    | Operational impact                                     |
|--------------------------|----------|--------------------------------------------------------|
| Remote command execution | Critical | Attacker can execute commands with elevated privileges |
| Credential compromise    | Critical | Passwords and hashes may support further access        |
| Phishing recurrence      | High     | Similar attachments can compromise additional users    |

## Recommendations

1. Isolate the host and block identified remote endpoints.
2. Remove malicious services only after collecting service configuration and binaries.
3. Reset affected credentials and review authentication logs for reuse.
4. Strengthen attachment filtering and user reporting procedures.

## Knowledge Demonstrated

- Service forensics
- Network attribution
- Browser artifact analysis
- Memory string analysis

## Conclusion

The combined evidence describes a complete attack chain from likely email delivery to execution, persistence, command-and-control and credential-oriented post-exploitation.

## Hands On Lab Practical 04

Completed as part of the authorized cyber security course practical work.